

2 An object of constant mass moves in a straight line. The variation with time t of the momentum p of the object is shown in Fig. 2.1.

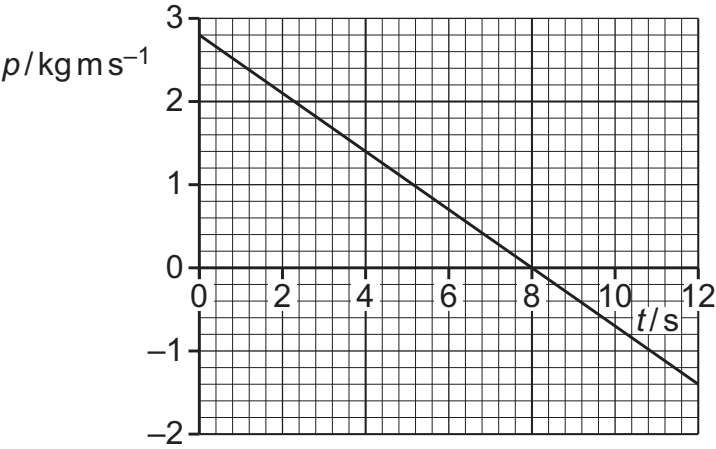


Fig. 2.1

(a) Define momentum.

.....
..... [1]

(b) Calculate the change in momentum of the object from time $t = 0$ to $t = 12\text{ s}$.

change in momentum = kg m s^{-1} [1]

(c) Calculate the magnitude of the resultant force acting on the object.

force = N [2]

(d) Describe the variation of the speed of the object from time $t = 0$ to $t = 8.0\text{ s}$.

.....
..... [1]

(e) By reference to Fig. 2.1, explain why the resultant force acting on the object during the first 8.0 s of its motion cannot be due to air resistance.

.....
.....
.....
..... [2]

(f) At time $t = 0$ the displacement d of the object is zero.

On Fig. 2.2, sketch the variation of d with time t from $t = 0$ to $t = 12\text{ s}$.

Numerical values of d are not required.

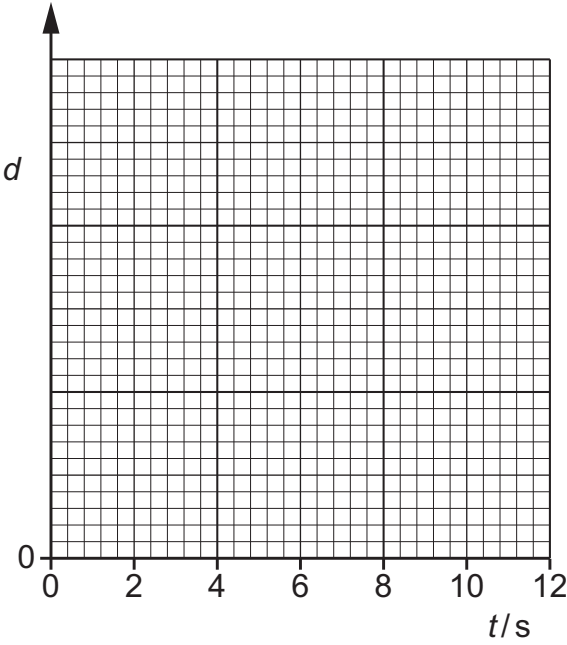


Fig. 2.2

[3]